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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TECHNICAL PRESENTATION

INTERNET OF THINGS(IOT) AND CLOUD COMPUTING : BUILDING CONNECTED ECOSYSTEMS

Team no. 10

CSA15 – Cloud Computing and Big Data Analytics

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Abstract

- Connected devices generate continuous data
- Cloud platforms provide scalable storage and processing
- Layered architecture links sensing, communication, analytics, and applications
- Focus: architecture, workflow, benefits, and design challenges

Aim and Objective

Aim:

Examine IoT–cloud architecture for connected ecosystems

Objectives:

- Identify IoT and cloud architecture layers
- Explain device-to-cloud data flow
- Understand storage, processing, and analytics
- Highlight security, scalability, and real-world use

Introduction

- Internet of Things (IoT): network of connected sensors, devices, and actuators
- Cloud computing: on-demand storage, computing, and services over the internet
- IoT + cloud: collect, transmit, process, and visualize device data
- Enables remote monitoring, automation, and data-driven decisions
- Architecture focus: device, edge/gateway, network, cloud platform, application

Core Technical Explanation

Problem Definition

- Large numbers of IoT devices generate continuous, heterogeneous data
- Device-only processing can be limited by compute, storage, and power
- Cloud integration introduces network dependence, latency, and security concerns
- Need: a layered architecture for reliable, scalable, secure device-to-cloud communication



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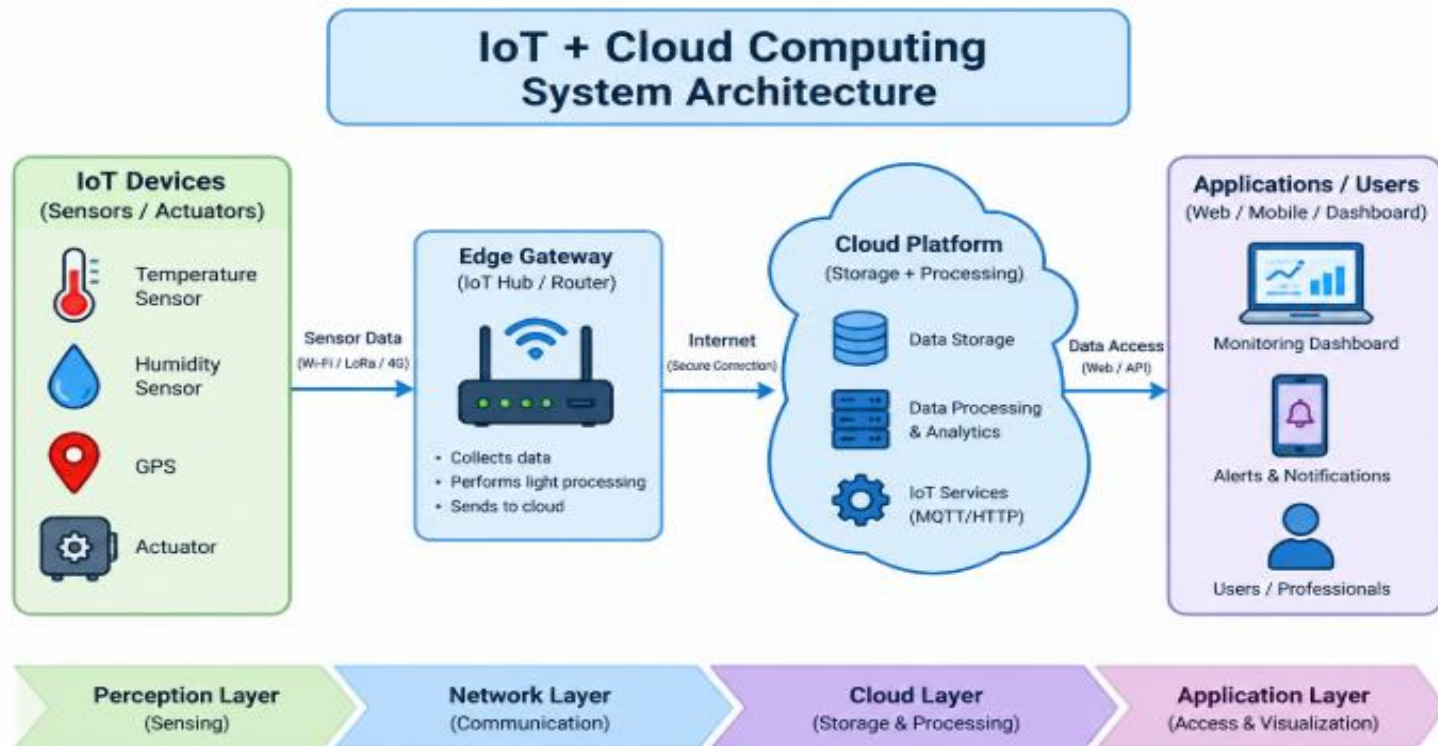
Existing System

- Standalone / device-centric monitoring
- Local storage and limited processing capacity
- Manual or periodic data collection
- Limited remote access and cross-device integration
- Scaling requires additional local infrastructure
- Potential gaps: interoperability, centralized analytics, and continuous visibility

Proposed System

- IoT sensors and devices capture real-time readings
- Gateway / edge layer filters and forwards data
- Secure network communication to cloud services
- Cloud ingestion, storage, processing, and analytics
- Web/mobile dashboard for monitoring and alerts
- Device management, access control, and data protection

Overall System Architecture Design



Proposed Methodology and Algorithm

Method: layered IoT–cloud data pipeline

- 1. Sense — collect measurements from IoT devices
- 2. Validate — check format, range, and device identity
- 3. Pre-process — filter noise and aggregate at edge (when suitable)
- 4. Transmit — send securely using MQTT or HTTPS
- 5. Ingest & store — accept data and persist it in cloud storage
- 6. Analyze — apply rules / analytics to detect events
- 7. Act — update dashboard, trigger alerts, or send control commands

Real-world Use Case in Cloud Computing

Smart agriculture: connected soil-moisture and weather sensors

Real-time entity: farm sensor network with cloud dashboard

- Sensors report soil moisture, temperature, and humidity
- Gateway forwards readings to a cloud IoT platform
- Cloud stores trends and compares readings with thresholds
- Dashboard supports remote irrigation monitoring
- Alerts can notify users when moisture falls below a set level

Illustrative use case; no specific deployment or measured results claimed



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Conclusion

- IoT devices provide continuous sensing and actuation
- Cloud services add scalable storage, processing, and remote access
- Layered architecture clarifies data movement from devices to applications
- Edge processing can reduce unnecessary traffic and response delay
- Security, interoperability, and connectivity remain key design concerns
- Connected ecosystems support monitoring, automation, and informed decisions

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